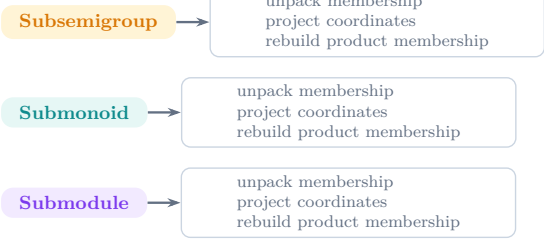


One proof, hiding in three places

PROOF REUSE

The repeated local pattern

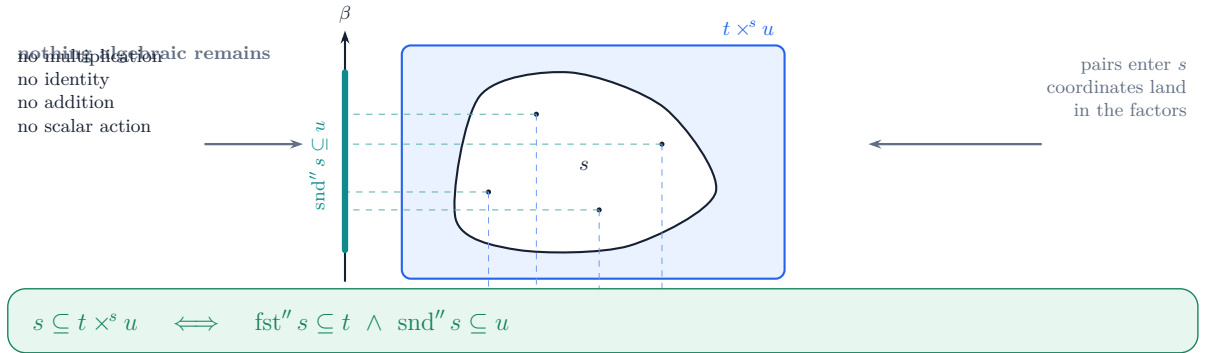
same element chase $\times$ the same ADL



signal: duplicated proof shape

Erase structure. Keep the set geometry.

weakest common statement



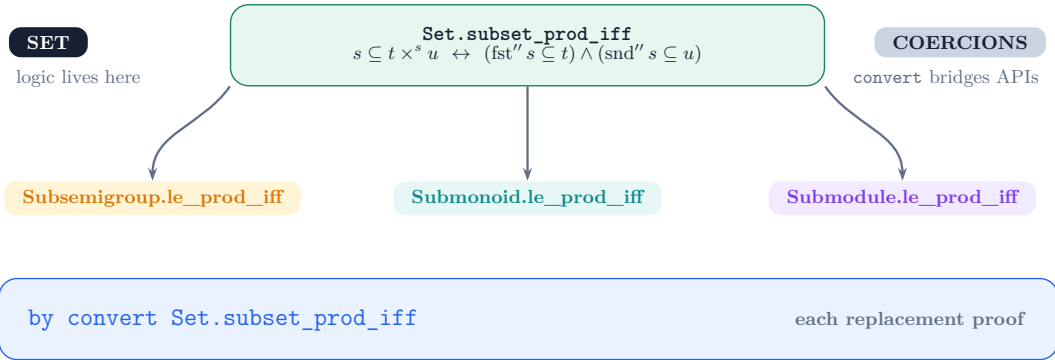
BIG IDEA

When proofs repeat below different structures, descend to the common carrier.

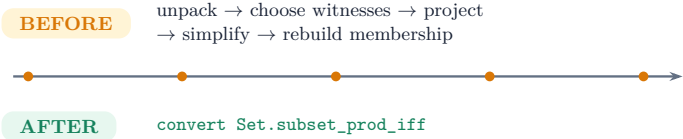
The generic lemma becomes a proof junction

PROOF REUSE

Three bundled theorems, one set-level engine



Proof compression



Why this abstraction boundary?

- WEAK** only sets, images, products
- NAMED** predictable beside `Set.subset_prod`
- REUSED** three existing proofs collapse



duplication ⇒ upstream-sized lemma